

FINAL SCHEME
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Total Pages: 3																																						
Scheme of Valuation/Answer Key (Scheme of evaluation (marks in brackets) and answers of problems/key)																																						
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FOURTH SEMESTER B.TECH DEGREE (R,S) EXAMINATION, JUNE 2023 (2019 SCHEME)																																						
Course Code: EET206																																						
Course Name: DIGITAL ELECTRONICS																																						
Max. Marks: 100		Duration: 3 Hours																																				
PART A																																						
<i>(Answer all questions; each question carries 3 marks)</i>																																						
1		Convert 1011 0010 to hexadecimal and Octal Hexadecimal – $(B2)_{16}$ Octal – 010 110 010 - $(262)_8$																																				
2		Draw the truth table and explain the operation of EX-OR and EX-NOR gates Truth table and expression of XOR gate <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th colspan="2">Input</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>Y</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> Truth table and expression of XNOR gate <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th colspan="2">Input</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>Y</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </tbody> </table>	Input		Output	A	B	Y	0	0	0	0	1	1	1	0	1	1	1	0	Input		Output	A	B	Y	0	0	1	0	1	0	1	0	0	1	1	1
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3		Get the dual of the expression $AB + \overline{A}C + A\overline{B}C$ $(AB + C) = 1$ Dual = $(A + B)(\overline{A} + \overline{C})[(A + \overline{B} + C) + (A + B)C] = 0$																																				
4		Derive the expression for sum and carry of a full adder																																				

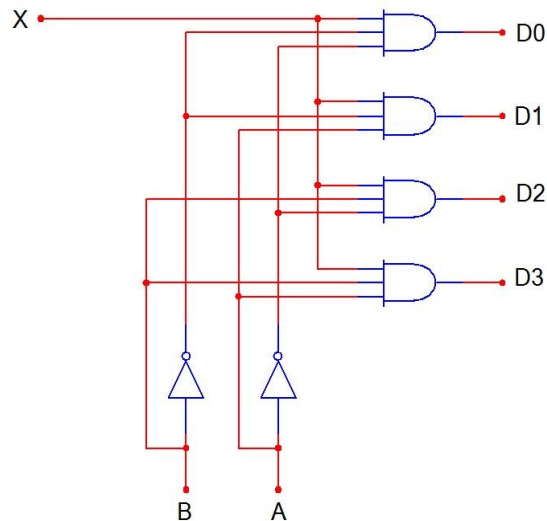
FINAL SCHEME

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	<table><tr><th colspan="3">Input</th><th colspan="2">Output</th></tr><tr><th>A</th><th>B</th><th>Cin</th><th>Sum</th><th>Carry</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> SUM = A XOR B XOR C ----- $C_{out} = AB + (A \oplus B)C_{in}$	Input			Output		A	B	Cin	Sum	Carry	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	1	1	1	0	0	1	1	1	1	1	1																																																																																			
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5	Explain a two bit comparator <table><tr><th colspan="4">INPUT</th><th colspan="3">OUTPUT</th></tr><tr><th>A₁</th><th>A₀</th><th>B₁</th><th>B₀</th><th>Y₁=A<B</th><th>Y₂=(A=B)</th><th>Y₃=A>B</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	INPUT				OUTPUT			A ₁	A ₀	B ₁	B ₀	Y ₁ =A<B	Y ₂ =(A=B)	Y ₃ =A>B	0	0	0	0	0	1	0	0	0	0	1	1	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	0	0	0	0	1	0	1	0	1	0	1	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	1	0	0	0	0	0	1	1	0	0	1	0	0	1	1	0	1	0	1	1	0	1	0	1	1	1	0	0	1	1	0	0	0	0	1	1	1	0	1	0	1	0	1	1	1	0	0	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	0
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6	Draw the circuit of a 1 to 4 Demultiplexer and explain																																																																																																																																					

FINAL SCHEME

02000EET206052102



- As in figure of 1-line-to-4-line demultiplexer (DEMUX) circuit, The data-input line goes to all of the AND gates.
- The two data-select lines enable only one gate at a time, and the data appearing on the data-input line will pass through the selected gate to the associated data-output line.
- In a demultiplexer, data are switched from one line to several lines

7

Explain the truth table and excitation table of a JK flip flop

$Q_n, J/K$	0	1
00	0	0
01	1	1
11	1	0
10	1	0

$$Q_{n+1} = Q_n K' + Q_n' J$$

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

8

What are the asynchronous inputs of a flip flop? Why are they called so?

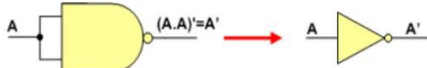
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02000EET206052102

		Preset and Clear - The normal data inputs of flipflop (D, T,S and R or J and K) are called synchronous inputs since they have effect on the outputs of the flip flop when the clocked pulse is present. - Flip flop can also have other inputs called Preset (or SET) and clear that can be used for setting the flip flop to 1 or resetting it to 0 by applying the appropriate signal to the Preset and Clear inputs. These inputs can change the state of the flip flop regardless of synchronous inputs or the clock. Both are active low signals generally. They are called asynchronous inputs or overriding inputs.
9		Draw the state diagram of a three bit binary down counter <pre> graph TD 111((111)) --> 110((110)) 110 --> 101((101)) 101 --> 100((100)) 100 --> 011((011)) 011 --> 010((010)) 010 --> 001((001)) 001 --> 000((000)) 000 --> 111 </pre>
10		What is FPGA? - The <u>field-programmable gate array (FPGA)</u> is an integrated circuit that consists of internal hardware blocks with user-programmable interconnects to customize operation for a specific application. The interconnects can readily be reprogrammed, allowing an FPGA to accommodate changes to a design or even support a new application during the lifetime of the part. - An FPGA is generally more complex and has a much higher density than a CPLD, although their applications can sometimes overlap. - As mentioned, the SPLD and the CPLD are closely related because the CPLD basically contains a number of SPLDs. - The FPGA, however, has a different internal structure (architecture), as illustrated

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02000EET206052102

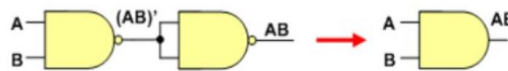
		in Figure. The three basic elements in an FPGA are the logic block, the programmable interconnections, and the input/output (I/O) blocks.
PART B		
<i>(Answer one full question from each module, each question carries 14 marks)</i>		
Module -1		
11	a)	Perform the following (i) 165.875_{10} to binary (ii) $4BAC_{16}$ to binary (iii) 378_{10} to octal (i) 165.875_{10} to binary 10100101.111_2 (ii) $4BAC_{16}$ to binary $0100\ 1011\ 1010\ 1100_2$ (iii) 378_{10} to octal 572_8
	b)	Universal Gates - NAND and NOR, Implementation of AND, OR, NOT using NAND The figure shows two ways in which a NAND gate can be used as an inverter (NOT gate). 1. All NAND input pins connect to the input signal A gives an output A'.  2. One NAND input pin is connected to the input signal A while all other input pins are connected to logic 1. The output will be A'. 

FINAL SCHEME

02000EET206052102

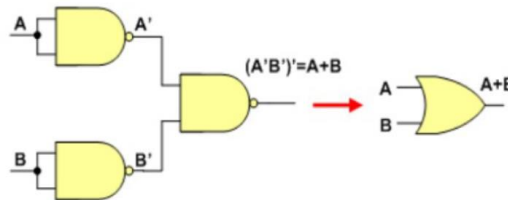
Implementing AND Using only NAND Gates

An **AND gate** can be replaced by NAND gates as shown in the figure (The AND is replaced by a NAND gate with its output complemented by a NAND gate inverter).



Implementing OR Using only NAND Gates

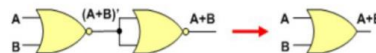
An **OR gate** can be replaced by NAND gates as shown in the figure (The OR gate is replaced by a NAND gate with all its inputs complemented by NAND gate inverters).



Implementation of AND, OR, NOT using NOR

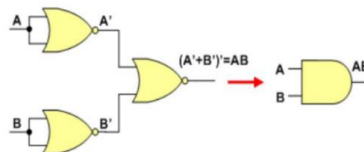
Implementing OR Using only NOR Gates

An **OR gate** can be replaced by NOR gates as shown in the figure (The OR is replaced by a NOR gate with its output complemented by a NOR gate inverter).



Implementing AND Using only NOR Gates

An **AND gate** can be replaced by NOR gates as shown in the figure (The AND gate is replaced by a NOR gate with all its inputs complemented by NOR gate inverters).



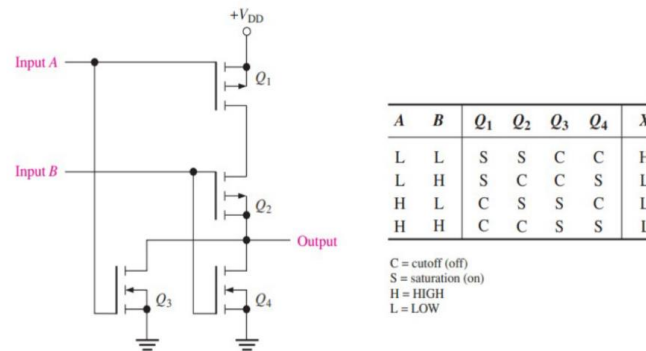
12 a) With the help of a neat diagram explain CMOS NAND gate

FINAL SCHEME

02000EET206052102

CMOS NOR gate – Internal Diagram

- Q1 and Q2 are P MOS transistors connected in series.
- Q3 and Q4 are N MOS transistors connected in parallel.



b) Describe error detection and correction using parity checking
Error detecting and correcting codes

- When a digital information is transmitted, it may not be received correctly by the receiver.
- The error is caused due to electrical disturbance of circuit it is also called noise.
- This noise may force '1' to change to '0' or vice versa.
- This error has to be detected and corrected.

There are two types of parity bits in error detection, they are

- Even parity
- Odd parity

Even Parity

- If the data has even number of 1's, the parity bit is 0. Ex: data is 10000001 -> parity bit 0
- Odd number of 1's, the parity bit is 1. Ex: data is 10010001 -> parity bit 1

Odd Parity

- If the data has odd number of 1's, the parity bit is 0. Ex: data is 10011101 -> parity bit 0
- Even number of 1's, the parity bit is 1. Ex: data is 10010101 -> parity bit 1

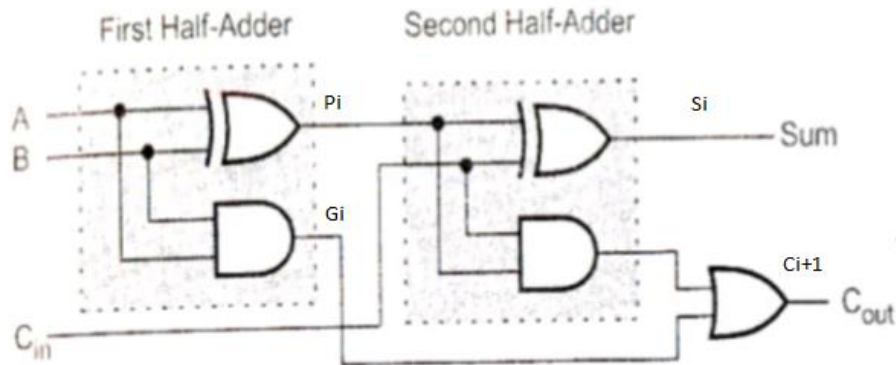
FINAL SCHEME
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Module -2																																																	
13	a)	Design and set up a half adder/subtractor circuit with mode control. Karnaugh Map for Sum and carry – 2 marks Sum/ Difference = 0.5 mark, $A'B + AB'$ (XOR GATE) Carry = 0.5 mark $M'AB + M A'B$ Circuit – 2 marks Truth Table - 2 marks <table><tr><th>M</th><th>A</th><th>B</th><th>Sum/Diff</th><th>Carry/Borrow</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>			M	A	B	Sum/Diff	Carry/Borrow	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	1	1	0	1	1	0	0	0	0	1	0	1	1	1	1	1	0	1	0	1	1	1	0	0
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	b)	Obtain the simplified POS expression using K- map – $Y = \sum m(1,3,7,11,15) + d(0,2,4)$ Hand-drawn Karnaugh map for a 4-variable function Y(A,B,C,D). The map is a 4x4 grid with rows labeled A,B (00, 01, 11, 10) and columns labeled C,D (00, 01, 11, 10). The cells contain values: (00,00)=0, (00,01)=1, (00,11)=1, (00,10)=0, (01,00)=1, (01,01)=0, (01,11)=1, (01,10)=0, (11,00)=0, (11,01)=0, (11,11)=1, (11,10)=1, (10,00)=0, (10,01)=0, (10,11)=1, (10,10)=0. There are four groups circled in red: a 2x2 square at (00,01), (01,01), (00,11), (01,11); a 2x2 square at (01,00), (11,00), (01,11), (11,11); a 2x2 square at (11,01), (11,11), (10,01), (10,11); and a 2x2 square at (11,11), (11,10), (10,11), (10,10). $= \bar{A}\bar{B} + CD$ Conversion of given SOP expression to POS – 2 marks K- Map – 1 marks, Grouping and Simplification – 2 marks, Boolean expression -2 marks																																															
14	a)	Using a neat circuit explain a 3 bit carry look ahead adder - The speed with which an addition can be performed is limited by the time required for the carries to propagate, or ripple, through all the stages of a parallel adder. - One method of speeding up the addition process by eliminating this ripple carry delay is called look-ahead carry addition. - The look-ahead carry adder anticipates the output carry of each stage, and based on																																															

FINAL SCHEME

02000EET206052102

the inputs, produces the output carry by either carry generation or carry propagation.



- Consider the full adder circuit shown above with corresponding truth table. If we define two variables as carry generate G_i and carry propagate P_i then,
 - $P_i = A_i \oplus B_i$
 - $G_i = A_i B_i$
- The sum output and carry output can be expressed as
 - $S_i = P_i \oplus C_i$
 - $C_{i+1} = G_i + P_i C_i$
- Where G_i is a carry generate which produces the carry when both A_i , B_i are one regardless of the input carry. P_i is a carry propagate and it is associate with the propagation of carry from C_i to C_{i+1} .
- The carry output Boolean function of each stage in a 4 stage carry-Lookahead adder can be expressed as

$$C_2 = G_1 + P_1 C_1$$

$$C_3 = G_2 + P_2 C_2$$

$$= G_2 + P_2 (G_1 + P_1 C_1)$$

$$= G_2 + P_2 G_1 + P_2 P_1 C_1$$

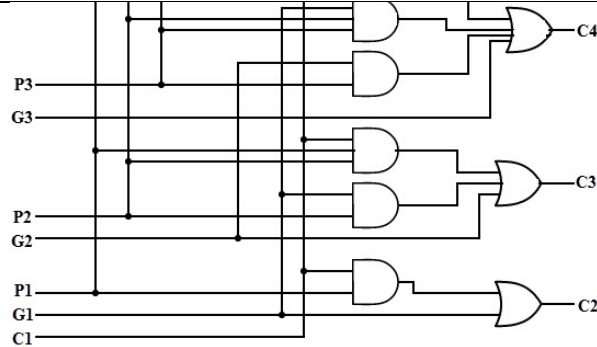
$$C_4 = G_3 + P_3 C_3$$

$$= G_3 + P_3 (G_2 + P_2 G_1 + P_2 P_1 C_1)$$

$$= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_1$$

FINAL SCHEME

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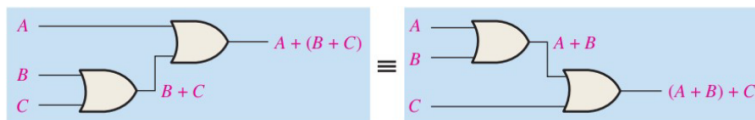
- b) Write and explain any five basic laws of Boolean algebra

Commutative Laws of addition

- The commutative law of addition for two variables is written as $A + B = B + A$
- This law states that the order in which the variables are OR ed makes no difference.

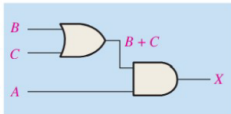
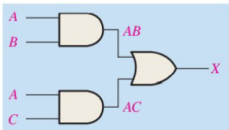
Associative law of addition

- The associative law of addition is written as follows for three variables
- $$A + (B + C) = (A + B) + C$$
- This law states that when ORing more than two variables, the result is the same regardless of the grouping of the variables.



FINAL SCHEME

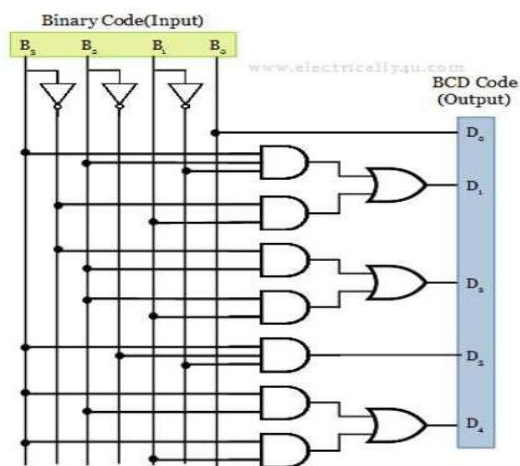
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		follows $A(B + C) = AB + AC$ • This law states that ORing two or more variables and then ANDing the result with a single variable is equivalent to ANDing the single variable with each of the two or more variables and then ORing the products. • The distributive law also expresses the process of factoring in which the common variable A is factored out of the product terms.  $X = A(B + C)$ $\equiv$  $X = AB + AC$																																																																																																														
Module -3																																																																																																																
15	a)	Implement $f(x,y,z) = \sum m(1,2,6,7)$ using 4 x 1 MUX Select lines = x, y Truth table , Circuit showing $D_0 = z$, $D_1 = \bar{z}$, $D_2 = 0$, $D_3 = 1$																																																																																																														
	b)	Explain a 3 to 8 decoder <table><tr><th colspan="3">Inputs</th><th colspan="8">Outputs</th></tr><tr><th>x</th><th>y</th><th>z</th><th>D0</th><th>D1</th><th>D2</th><th>D3</th><th>D4</th><th>D5</th><th>D6</th><th>D7</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table>	Inputs			Outputs								x	y	z	D0	D1	D2	D3	D4	D5	D6	D7	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0	0	0	0	0	1
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16	a)	Design and set up a Binary to BCD converter circuit																																																																																																														

FINAL SCHEME

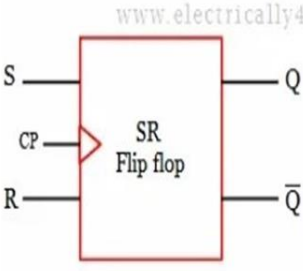
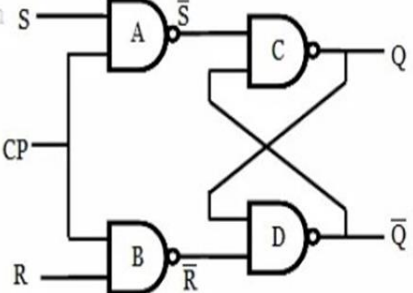
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Decimal Number	Binary code (Input)				BCD code (Output)				
	B ₃	B ₂	B ₁	B ₀	D ₄	D ₃	D ₂	D ₁	D ₀
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	0
3	0	0	1	1	0	0	0	1	1
4	0	1	0	0	0	0	1	0	0
5	0	1	0	1	0	0	1	0	1
6	0	1	1	0	0	0	1	1	0
7	0	1	1	1	0	0	1	1	1
8	1	0	0	0	0	1	0	0	0
9	1	0	0	1	0	1	0	0	1
10	1	0	1	0	1	0	0	0	0
11	1	0	1	1	1	0	0	0	1
12	1	1	0	0	1	0	0	1	0
13	1	1	0	1	1	0	0	1	1
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15	1	1	1	1	1	0	1	0	1



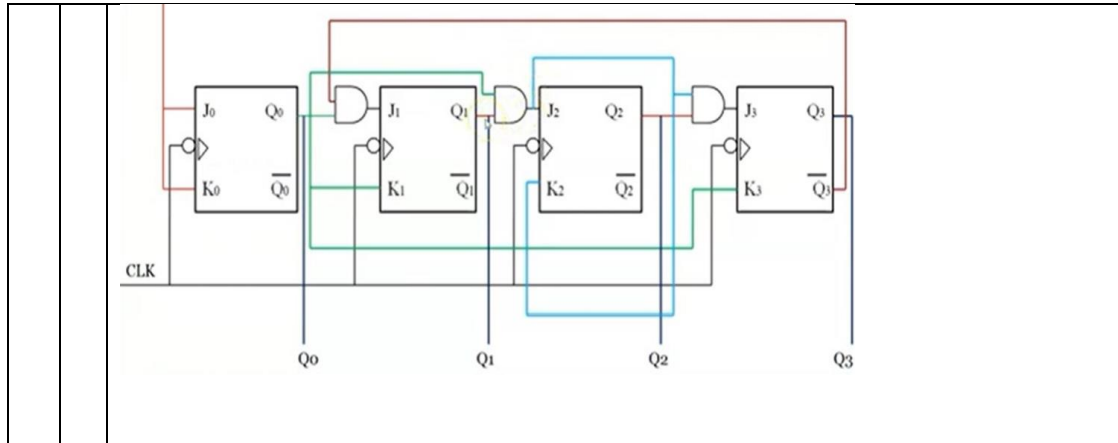
FINAL SCHEME

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	b)	Explanation of priority encoder • Priority encoder performs the same basic encoding function which encode the information from 2^n bit lines into n bit code. • A priority encoder also offers additional flexibility in that it can be used in applications that require priority detection. • The priority function means that the encoder will produce a BCD output corresponding to the highest-order decimal digit input that is active and will ignore any other lower-order active inputs. • As the name indicates, the priority is given to inputs line. • If two or more input lines are high at the same time i.e 1 at the same time, then the input line with high priority shall be considered. • For instance, if the 6 and the 3 inputs are both active, the BCD output is 0110 (which represents decimal 6).
Module -4		
17	a)	Draw the logic circuit of SR flip flop and explain it • Basic circuit of SR latch can be modified by providing an additional control input when state is to be changed. • It is constructed using NAND latch. • If clock pulse is zero flipflop will be inactive and it will enter into Hold state. If clock pulse is applied that means it is one, flipflop will work according to the input.  
	b)	What is modulus of a counter. Draw the circuit of an asynchronous decade counter • A counter with n flip flops can have 2^n states. • The no of states in a counter is called modulus or mod of a counter. • No of flip flops in a counter determines its count. • A counter can count from 0 to $2^n - 1$ and vice versa.

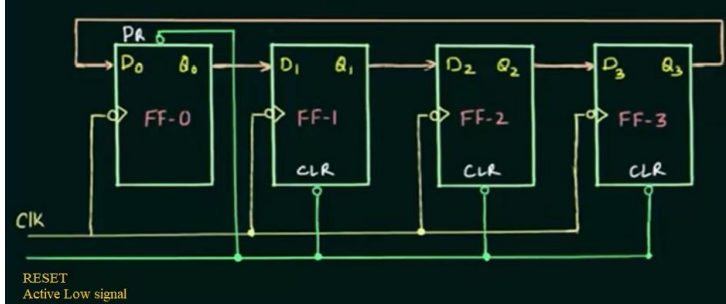
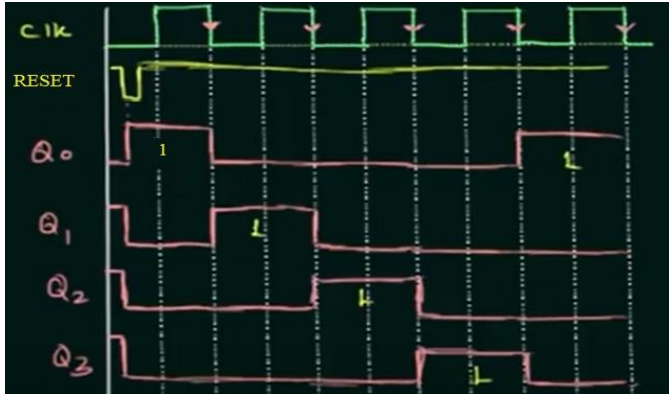
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FINAL SCHEME

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18	a) Enumerate the steps for design of a synchronous counter 1. Select Type ▪ Up or Down ▪ Modulus 2. Select Flip-Flop Type ▪ J-K or D ▪ Positive Edge Trigger (PET) or Negative Edge Trigger (NET) 3. Determine Number of Flip-Flops ▪ $(2^{\# \text{ Flip-Flop}} \geq \text{Modulus})$ 4. Design Basic Counters ▪ Same polarity for down counters: ▪ Opposite polarity for up counters: Design Limits Logic ▪ Input to logic is count that is one past the end of sequence.
	b) Draw the circuit of 4 bit ring counter and give the timing diagram  
Module -5	

FINAL SCHEME

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19	a)	Design a 4 bit weighted resistor DAC whose full scale output voltage is -5V. The logic levels are Logic 1 = +5 V and logic 0 = 0 V. What is the output voltage when the input is 1101 Design Equation – 3 marks Full scale output = Input value = 1111, Therefore $-(5 + (5/2) + (5/4) + (5/8)) \times (R_f/R) = -5,$ Let $R_f = 10 \text{ K ohm}$, $R = 18.75 \text{ K ohm}$, $2R = 37.5 \text{ K ohm}$, $4R = 75 \text{ K ohm}$, $8R = 150 \text{ K ohm}$ – 4 marks V out corresponding to 1101 = - 4.333 V – 3 marks															
	b)	Give the Verilog code for AND gate  <table border="1" style="margin: 10px auto;"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </tbody> </table> Circuit Diagram With TT <pre> module andgate (a, b, y); input a, b; output y; assign y = a & b; endmodule </pre>	A	B	Output	0	0	0	0	1	0	1	0	0	1	1	1
A	B	Output															
0	0	0															
0	1	0															
1	0	0															
1	1	1															
20	a)	Explain Flash type A/D converter • A simultaneous or flash type ADC is based on comparing an analog input voltage with a set of reference voltages . • It is also known as parallel ADC. • It is the fastest ADC and less complex compared to others. • An n bit flash ADC uses op-amp comparators and resistors. • For e.g., a 2-bit Flash type ADC requires 3 comparators and 4 resistances to divide the voltage.															

FINAL SCHEME

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	Resolution $\Delta V = \frac{V_{ref}}{2^N} = \frac{8}{2^3} = 1.0 \text{ V}$ $2^3 - 1 = 7$ Comparators
b)	Differentiate PAL and PLA • The PAL is the opposite of the ROM, having a programmable set of ANDs combined with fixed ORs. • PAL is a programmable logic device that has Programmable AND array & fixed OR array. • The PAL structure allows any sum-of-products (SOP) logic expression with a defined number of variables to be implemented. • Here, the inputs of AND gates are programmable. That means each AND gate has both normal and complemented inputs of variables. So, based on the requirement, we can program any of those inputs. So, we can generate only the required product terms by using these AND gates. • Here, the inputs of OR gates are not of programmable type. So, the number of inputs to each OR gate will be of fixed type. Hence, apply those required product terms to each OR gate as inputs. Therefore, the outputs of PAL will be in the form of sum of products form *****